CSC 163: Computing, Ethics, and Society

Software Law: Copyright, Licensing, and Illegal Use

ABET SO4, performance indicator 4a

Roadmap

1. **Copyright basics for software**
2. **Software licenses**: proprietary, permissive, copyleft
3. **Illegal uses of software** and who is harmed
4. **Remedies**: civil, criminal, DMCA
5. **Cases and worked examples**
6. **How to answer a 4a-style question**

Learning Objectives (SO4, PI 4a)

After this lecture you can:

- State what copyright protects in software, and what it does not
- Explain the difference between owning a copy and owning the copyright
- Compare proprietary, permissive and copyleft licenses
- Describe **two distinct situations** where using software is illegal, explain **what makes each illegal**, and identify **who is harmed**
- Name the main civil and criminal consequences

ABET SO4: recognize professional responsibilities and make informed judgments in computing practice based on legal and ethical principles.

Copyright Basics

What the law protects when you write software

What Copyright Protects in Software

- **Protected:** the original expression in a program: source code and object code (the Copyright Office: all copyrightable expression embodied in the program)
- **Not protected:** ideas, algorithms, functions, logic, formatting and system design
- Protection is **automatic** once the code is fixed in a tangible form. Registration is optional, but a U.S. work generally must be registered before the owner can sue
- Example: anyone may write a program that uses quicksort. The algorithm is free; each program's code is protected

Source: U.S. Copyright Office, Circular 61 (computer programs)

The Owner's Exclusive Rights

Only the copyright owner (or someone with permission) may:

- **Reproduce** the program: copy it, install copies
- **Distribute** copies: sell, share, upload
- **Prepare derivative works**: modify, translate, port to another platform
- **Publicly perform or display** the work

Everyone else needs a **license** (permission), or an exception such as fair use

Owning a Copy vs Owning the Copyright

- Most software is **licensed, not sold**: you get a right to use it under stated terms
- A physical book can be resold (first-sale doctrine). Courts have often treated software as licensed, so first sale may not apply (*Vernor v. Autodesk*, 9th Cir. 2010)
- The lawful owner of a copy may make a backup copy or a copy that is an essential step in using it (17 U.S.C. 117)
- **Who owns code you write?** An employee's code within the scope of the job is usually a work made for hire, owned by the employer. Contractors and students: check the contract or university policy

Software Licenses

Permission with conditions

Types of Software Licenses

Type	Examples	What you may do	Key condition
Proprietary (EULA)	Commercial apps	Use as the license allows	No source code; no copying, redistribution or reverse engineering unless allowed
Free trial / freeware	Trials, free tiers	Use free, often limited	Not open source; may be time-limited or non-commercial
Permissive open source	MIT, Apache 2.0	Use, modify, redistribute, even in closed products	Keep copyright and license notices
Weak copyleft	LGPL	Link to the library from your own code	Changes to the library itself stay under the same terms
Strong copyleft	GPL, AGPL	Use, modify, redistribute	Larger works must be released under the same license, with source
No license	Public repo with no LICENSE file	View or fork as the host allows	All rights reserved: reuse needs the author's permission

Permissive vs Copyleft

- **Permissive (MIT, Apache 2.0):** main duty is to keep notices. Your larger work may use different terms and need not release source. Apache 2.0 adds a patent grant from contributors
- **GPLv3:** source for the licensed work and for larger works that use it must be offered under the same license
- **LGPLv3:** the library stays open; code that merely uses it through its interfaces can have other terms
- **AGPLv3:** like GPL, and also applies when a modified version runs as a **network service**
- **Key idea:** a license is a set of conditions. Breaking the conditions removes the permission

Source: choosealicense.com

Typical Restrictions in Licenses

- **Scope of use:** number of users, seats or devices; personal vs commercial; student or educational only
- **Copying and sharing:** no redistribution, no resale, no uploading to public sites
- **Reverse engineering and modification:** often banned in proprietary licenses
- **Duration and platform:** subscription term, trial period, cloud or on-premise limits
- **Attribution and share-alike:** notices to keep (permissive), source to share (copyleft)

Reading the license is a professional habit: it tells you what you may do

Illegal Uses of Software

What makes a use illegal, and who pays for it

Illegal Use: Piracy and License Violations

Situation	What makes it illegal	Who is harmed
1. Piracy: sharing or selling copies of paid software without permission	Copying and distributing without permission infringes the owner's exclusive rights (17 U.S.C. 106, 501)	Developers and publishers lose sales; people who run pirated copies face malware risk
2. Over-installing: 50 installs with 20 purchased licenses	Breach of the license terms; copies beyond the license can also infringe	The vendor loses license revenue
3. Wrong scope: using a student license for paid client work	Use outside the licensed scope: breach of contract, possibly infringement	The vendor; the program that offered the discount

Illegal Use: Circumvention, Open Source, Access

Situation	What makes it illegal	Who is harmed
4. Circumvention: keygen or patch that bypasses license checks or DRM	DMCA section 1201 bans circumventing technical measures that control access to a work	The vendor; users of cracks, who risk malware
5. Open-source violation: GPL code shipped in a closed-source product	Violating license conditions is copyright infringement (*Jacobsen v. Katzer*, Fed. Cir. 2008)	The original authors; downstream users who lose access to source
6. Unauthorized access: using a tool to log in to accounts that are not yours	Computer Fraud and Abuse Act, 18 U.S.C. 1030: access without authorization	System owners and the people whose data is exposed

Remedies and Consequences

Civil

- Court orders to stop
- Actual damages plus the infringer's profits, **or** statutory damages of **$750 to $30,000 per work**
- Up to **$150,000 per work** if willful; as low as **$200** if innocent

Criminal

- Willful infringement is a crime if done for commercial advantage or private gain
- Or if copies worth **over $1,000** are made or distributed in 180 days
- Fines and prison under federal law

Platform and contract

- DMCA takedown notices to hosts
- Account termination for repeat infringers
- Breach-of-contract claims under the license

Sources: 17 U.S.C. 504 and 506; DMCA summary (Copyright Office)

The DMCA in Brief

- **Section 1201 (anti-circumvention):** it is illegal to circumvent technical measures that control **access** to a copyrighted work, and to sell tools made for that purpose. Exceptions include security testing, encryption research and some reverse engineering
- **Section 512 (safe harbors):** online services avoid damages if they remove infringing material promptly after a proper notice
- **Notice and takedown:** owner sends a notice; the user may send a counter-notice; the material returns in 10 to 14 business days unless the owner sues
- Providers must have a **repeat-infringer** policy

Source: U.S. Copyright Office, DMCA summary

Who Is Harmed, and How

- **Developers and publishers:** lost revenue, less money for maintenance and new products
- **Open-source authors and communities:** conditions ignored, contributions taken without the promised sharing
- **Honest users:** exposure to malware in cracked software; less trust in the ecosystem
- **Society:** weaker incentives to create software, and weaker respect for rules that everyone relies on
- **Balance:** critics note that strict DRM and anti-circumvention rules can also burden legitimate users and researchers. A good answer names the harmed party **and** how they are harmed

Two Cases to Know

Vernor v. Autodesk (9th Cir. 2010)

- Vernor resold used copies of design software
- Court: buyers were **licensees**, not owners, so first sale did not apply
- Lesson: read what the license says about transfer

Jacobsen v. Katzer (Fed. Cir. 2008)

- Open-source license terms (attribution and notice of changes) were ignored
- Court: these terms are enforceable **conditions** of the copyright license
- Lesson: free software still has legal rules

Case summaries for teaching, not legal advice. Check the opinions before relying on them.

Worked Example: Applying the Three Questions

	A company installs 50 copies of a design tool, buys 20 seats	**A startup ships a GPL library inside a closed app**
What happened	Installs exceed the seats purchased	Source code of the app is not offered; GPL notice removed
Why illegal	Breach of the license scope; extra copies are not authorized	GPL conditions broken, so no permission to distribute; infringement
Who is harmed	The vendor, who loses license revenue	The library's authors, and users who cannot get the source

How to Answer a 4a-Style Question

1. Situation	Name a concrete case: piracy, over-installing, circumvention, GPL violation, unauthorized access
2. What makes it illegal	Cite the rule: copyright (copying, distribution), a license term, DMCA 1201, CFAA
3. Who is harmed	Name the party and how: lost revenue, malware, lost access to source, exposed data
4. Be distinct	A second situation must differ in kind (for example piracy vs license violation)

Level 4 answers use precise legal bases and discuss the nature of the harm.

Summary

- Copyright protects **expression** in code, automatically; ideas and algorithms are free to use
- Most software is **licensed**: conditions define what is legal
- Illegal uses include **piracy, over-installing, wrong-scope use, circumvention, open-source violations and unauthorized access**
- Consequences: injunctions, damages (statutory up to $150,000 per work if willful), criminal penalties, takedowns
- Always answer in three parts: **situation, legal basis, harmed party**

Exercises and Solutions

Practice for performance indicator 4a

Exercise 1: Is it illegal?

Decide whether each use is legal or illegal, and name the main reason.

Scenario	Legal or illegal?	Main reason
Uploading a full copy of a paid game to a file-sharing site		
A company installs 50 copies of a program on a 20-seat license		
Using a keygen to unlock a paid program		
Shipping a GPLv3 library inside a closed-source app, source withheld		
Using an MIT-licensed library in a closed product and keeping its license notice		
Making a backup copy of software you lawfully own or license, for your own use		

Solution 1: Is it illegal?

Check the rule behind each case: copying and distribution rights, the license conditions, anti-circumvention (DMCA 1201).

Scenario	Legal or illegal?	Main reason
Uploading a full copy of a paid game to a file-sharing site	**Illegal**	**Unauthorized reproduction and distribution (copyright infringement)**
A company installs 50 copies of a program on a 20-seat license	**Illegal**	**Breach of the license scope; copies beyond the license are unauthorized**
Using a keygen to unlock a paid program	**Illegal**	**Circumvention of access controls (DMCA 1201) and an unlicensed copy**
Shipping a GPLv3 library inside a closed-source app, source withheld	**Illegal**	**GPL conditions broken, so no permission to distribute**
Using an MIT-licensed library in a closed product and keeping its license notice	**Legal**	**MIT is permissive; the notice condition is met**
Making a backup copy of software you lawfully own or license, for your own use	**Generally legal**	**Allowed for the lawful owner of a copy (17 U.S.C. 117), if the license permits**

Exercise 2: Match the license

Match each license to its key condition.

License	Key condition
MIT	
Apache 2.0	
GPLv3	
LGPLv3	
AGPLv3	
Proprietary EULA	

Solution 2: Match the license

Permissive licenses ask mostly for notices. Copyleft licenses require sharing source of larger works. Proprietary licenses restrict copying and modification.

License	Key condition
MIT	**Keep the copyright and license notice; use freely, even in closed products**
Apache 2.0	**Keep notices and state changes; includes a patent grant from contributors**
GPLv3	**Larger works that use it must be released under GPL with source**
LGPLv3	**The library stays open; code that only links to it can have other terms**
AGPLv3	**Like GPL, and also applies when modified code runs as a network service**
Proprietary EULA	**Use only as the license allows; no copying, resale or reverse engineering unless granted**

Exercise 3: Statutory damages

A court finds that a website distributed pirated copies of 4 different programs (4 works). Statutory damages are awarded per work: $750 to $30,000 normally; up to $150,000 if willful; as low as $200 if innocent. Compute the total for each case.

Case	Per work	Total for 4 works
Ordinary minimum	$750	
Ordinary maximum	$30,000	
Willful maximum	$150,000	
Innocent minimum	$200	

Solution 3: Statutory damages

Multiply the per-work amount by the number of works.

Note: the award is per work infringed, not per copy. If the site had distributed 1 program 4 times, the number of works would be 1, and the ordinary minimum would be **$750**.

Case	Per work	Total for 4 works
Ordinary minimum	$750	**$3,000**
Ordinary maximum	$30,000	**$120,000**
Willful maximum	$150,000	**$600,000**
Innocent minimum	$200	**$800**

Exercise 4: True or false

Write T or F. If false, correct it with a few words.

Statement	T / F	Correction (if false)
Copyright protects the algorithm used in a program		
Most software is licensed, not sold		
Open-source software has no legal conditions		
Willful infringement can bring up to $150,000 per work		
Copyright must be registered before the work is protected		
Cracking a license check is illegal only if you sell the crack		

Solution 4: True or false

1 point for T/F, 1 point for the correction.

Statement	T / F	Correction (if false)
Copyright protects the algorithm used in a program	F	It protects expression (code), not ideas or algorithms
Most software is licensed, not sold	T	—
Open-source software has no legal conditions	F	Licenses set conditions; breaking them removes permission
Willful infringement can bring up to $150,000 per work	T	—
Copyright must be registered before the work is protected	F	Protection is automatic; registration is needed before suing (U.S. works)
Cracking a license check is illegal only if you sell the crack	F	Circumventing access controls is itself prohibited (DMCA 1201)

Exercise 5: Practice 4a question

Describe **two distinct situations** where using software is illegal.
For each, explain **what makes it illegal** and **who is harmed**.
(Target: about half a page.)

Solution 5: Practice 4a question

- **Model answer.**
- Situation 1, piracy: a student shares a paid design program with friends through a file-sharing site. Copying and distributing without permission infringes the owner's exclusive rights (17 U.S.C. 106 and 501). The developer loses sales; users of the copies risk malware.
- Situation 2, license violation: a company buys 20 licenses and installs 50 copies. The license limits use to 20 seats, so the extra copies are unauthorized (breach of contract and infringement). The vendor loses revenue that funds maintenance.
- **Scoring (4 points):** two situations that differ in kind (1); correct legal basis for each (1); harmed party named (1); how they are harmed, with consequences such as damages up to $150,000 per work if willful (1).

References

- U.S. Copyright Office, Circular 61: Copyright Registration of Computer Programs. copyright.gov/circs/circ61.pdf
- 17 U.S.C. chapter 5 (sections 504 and 506). copyright.gov/title17/92chap5.html
- U.S. Copyright Office, The Digital Millennium Copyright Act of 1998: summary. copyright.gov/legislation/dmca.pdf
- Choose a License: licenses overview. choosealicense.com/licenses/
- Vernor v. Autodesk, Inc., 621 F.3d 1102 (9th Cir. 2010); Jacobsen v. Katzer, 535 F.3d 1373 (Fed. Cir. 2008)
- ABET, Criteria for Accrediting Computing Programs (student outcome 4). abet.org
- These slides are original teaching summaries, not legal advice.